Research Idea: NGC 4429

ALMA Thesis Planner (automated pipeline)

Title: Testing the "true cavity vs. tracer-faint" hypothesis for the NGC 4429 circumnuclear hole using dense-gas HCN(1-0) and Band 3 continuum imaging

Description: This thesis uses only the two products already delivered for ALMA project 2023.1.01214.S — the pipeline-calibrated HCN(1-0) spectral cube (320×320 pixels, 126 channels, ~49.6 km/s channel width, beam $\approx 1.88'' \times 1.81''$, PA ~80°) and the Band 3 continuum image (beam $\approx 1.63'' \times 1.55''$, PA ~80°, combining spw 17/21/23/25 at ~86.4/88.3/98.4 GHz) — to characterize dense molecular gas and continuum emission in NGC 4429's nuclear region and test whether HCN(1-0), a tracer of much denser gas (n_crit ~10^6 cm^-3) than CO(2-1), shows the same central deficit reported for CO(2-1) in the proposal's background text.

Before committing to the full analysis, the student will first do a quick feasibility check: measure the cube's channel-by-channel RMS noise directly from line-free channels to confirm HCN(1-0) is plausibly detectable at all in this nucleus (early-type galaxies are frequently dense-gas-poor, and no sensitivity figures are given in the data description, so this must be verified from the data itself rather than assumed).

The student will then: (1) produce moment 0/1/2 maps of the HCN(1-0) cube (line centered on the redshifted rest frequency 88.6316 GHz) across the ~60″ Band 3 primary beam, focusing on the nuclear ~10″; (2) extract a radial intensity profile centered on the SMBH position (RA 12:27:26.503, Dec +11:06:27.58) to check for a central intensity deficit or plateau in HCN, converting the beam FWHM to a physical scale using a literature-standard distance to NGC 4429 (to be selected and cited during the thesis, e.g. from a standard distance catalog) — since the previously-reported CO(2-1) hole (~10–100 pc) is comparable to or smaller than this beam, the thesis explicitly frames its result as a beam-limited test (detection/non-detection at the beam's physical resolution), not a re-measurement of the hole's true size or depth, which remains unverified background from the proposal text; (3) if no significant nuclear HCN is detected, compute a rigorous channel-by-channel noise-based upper limit on HCN(1-0) flux/column density at the nucleus, and convert it to a dense-gas mass upper limit using a literature-standard HCN-to-H2 conversion factor (to be selected and cited, with the choice explicitly flagged as an assumption), reporting the mass limit as a range under a sensitivity check across a small, explicitly bounded span of alternative literature conversion-factor values rather than a single derived number; (4) separately characterize the Band 3 continuum image for the presence/absence, location, and morphology of a compact nuclear source coincident with the SMBH position versus diffuse/extended emission, with an explicit, stated caveat that this single narrow-band (~86–98 GHz) continuum image cannot by itself distinguish AGN synchrotron/free-free emission from dust or stellar continuum, and is therefore reported only as a morphological/positional constraint, not an AGN diagnostic; (5) synthesize these into a discussion of whether the available evidence is more consistent with a genuinely gas-poor cavity (low HCN + no notable compact continuum) or an unresolved/tracer-dependent hole (HCN plausibly present below current resolution/sensitivity), tying this back to the proposal's AGN-feedback-vs-secular question while explicitly noting that a firm answer requires the CO(1-0)/CO(6-5) data not included in this member OUS's delivered products, which are out of scope here.

This remains a self-contained, single-cube-plus-single-image analysis using standard moment-map, radial-profile, and noise/upper-limit techniques (e.g., CASA or Python/spectral-cube), fully achievable by one student in a semester to a year, requiring no new observations, additional bands, or externally sourced datasets beyond the two FITS products already in hand plus one distance value and one HCN-to-H2 conversion factor, each to be selected from the literature and explicitly cited during the thesis, used solely for standard unit conversions rather than as substitute science data.